

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 11

Probability, Universality, and Levels of Description

Parallel Reading

Self-Study Curriculum

Parallel Reading: Terence Tao, “E pluribus unum: From Complexity, Universality” (*Daedalus*, 2012, approximately 15 pages, freely available online): a working mathematician’s essay on why wildly different systems share the same large-scale statistics. Read it first; it is the conceptual spine of this chapter.

Also read: E. T. Jaynes, *Probability Theory: The Logic of Science* (Cambridge, 2003), Chapter 1 (“Plausible Reasoning,” approximately 20 pages): the cleanest statement of the “probability as extended logic / degree of belief” position. (**Core reading $\approx$ 35 pages**, deliberately light, because the ideas are heavy.)